# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET D

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

**Time: 1 Hour 30 Minutes** Name: _____ Roll No.: _____ **Maximum Marks: 40**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper wherever a graph is asked.

## SECTION — A (Objective Type Questions) 10 × 1 = 10

*Choose the correct option and write it in your answer book.*

**1.** The lines $2x - y = 1$ and $4x - 2y = 2$ are: 1

**(a)** intersecting **(b)** perpendicular **(c)** coincident **(d)** parallel

**2.** The lines $x + y = 4$ and $x - y = 0$ intersect at: 1

**(a)** $(2, 2)$ **(b)** $(1, 3)$ **(c)** $(0, 4)$ **(d)** $(4, 0)$

**3.** $ax + 2y = 9$ and $3x + by = 18$ have infinitely many solutions when $(a, b)$ is: 1

**(a)** $(3, 4)$ **(b)** $(\frac{3}{2}, 2)$ **(c)** $(\frac{3}{2}, 4)$ **(d)** $(6, 1)$

**4.** A student states: "Every linear equation in two variables has a unique solution." The correct response is: 1

**(a)** It has two solutions

**(b)** Correct

**(c)** Incorrect — it has infinitely many solutions

**(d)** It has no solution

**5.** "2 pens and 3 pencils cost ₹40" ($x$ = pen, $y$ = pencil): Student A writes $2x + 3y = 40$; Student B writes $3x + 2y = 40$. Then: 1

**(a)** Student B is correct **(b)** Student A is correct

**(c)** Both are correct **(d)** Both are wrong

**6.** The roots of $3x^2 - 27 = 0$ are: 1

**(a)** $\pm 9$ **(b)** $\pm\sqrt{3}$ **(c)** $\pm 3$ **(d)** 3 only

**7.** If $x = -\frac{1}{2}$ is a root of $3x^2 + 2kx - 3 = 0$, then $k$ is: 1

**(a)** $-\frac{4}{9}$ **(b)** $-\frac{9}{4}$ **(c)** $\frac{9}{4}$ **(d)** 2

**8.** The discriminant of $2x^2 - 7x + 6 = 0$ is: 1

**(a)** 49 **(b)** −1 **(c)** 97 **(d)** 1

**9.** A student states: "The only consecutive integers whose product is 30 are 5 and 6." The correct response is: 1

**(a)** Only $-5, -6$ work **(b)** Correct

**(c)** Incorrect — $-6$ and $-5$ also work **(d)** None exist

**10.** Which of the following is **not** a quadratic equation? 1

**(a)** $(x + 2)^2 = x^2 + 3$ **(b)** $2x^2 + 1 = 0$ **(c)** $x^2 = 5x$ **(d)** $x(x + 1) = 6$

## SECTION — B (Very Short Answer Type) — 4 × 2 = 8

*Answer in about 20–30 words each.*

**11.** Solve by substitution: $s - t = 3$ and $\frac{s}{3} + \frac{t}{2} = 6$. 2

**12.** Is the pair $x + y = 5$ and $2x + 2y = 10$ consistent? Write any two of its solutions. 2

**13.** Solve for $x$: $x^2 - 2ax + a^2 - b^2 = 0$. 2

**14.** Find the roots of $2x^2 + x - 4 = 0$ using the quadratic formula. 2

## SECTION — C (Short Answer Type) — 4 × 3 = 12

*Answer in about 40–60 words each.*

**15.** If 3 students shift from section A to section B, both sections have equal strength. If 3 students shift from B to A, section A has double the strength of B. Find the strength of each section. 3

**16.** Find two numbers whose difference is 4 and whose product is 96. 3

**17.** Solve graphically $3x + y - 11 = 0$ and $x - y - 1 = 0$. Shade the region bounded by these lines and the $y$-axis. 3

OR

A fraction becomes $\frac{4}{5}$ if 1 is added to both its numerator and denominator. If 5 is subtracted from both, it becomes $\frac{1}{2}$. Find the fraction. 3

**18.** The perimeter of a right triangle is 60 cm and its hypotenuse is 25 cm. Find the other two sides. 3

OR

Check whether $(x - 2)^2 + 1 = 2x - 3$ is a quadratic equation. If yes, solve it. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**19.** **Case study:** A salesperson gets a fixed monthly salary plus a commission on each item sold. Selling 40 items she earns ₹14,000; selling 60 items she earns ₹16,000. 5

(i) Form the equations.

(ii) Find the salary and commission per item.

(iii) Find her earnings for 75 items.

(iv) How many items must she sell to earn ₹20,000?

OR

The cost of 4 pens and 4 pencil boxes is ₹100. Three times the cost of a pen is ₹15 more than the cost of a pencil box. Form the equations and solve them graphically. Verify your answer. 5

**20.** **Case study:** A group of students hired a bus for a trip for ₹4800, sharing the cost equally. When 4 students dropped out, each remaining student had to pay ₹40 more. 5

(i) Form the equation.

(ii) Find the original number of students.

(iii) Find the original share of each student.

OR

A rectangular garden is 3 m longer than it is wide and its area is 180 m². 5

(i) Find its dimensions.

(ii) Find the cost of fencing it at ₹20 per m.

(iii) A student says the width is 15 m. Comment.

**END OF QUESTION PAPER** — **Total: 40 Marks**